

Research-to-operational hourly PM_{2.5} forecasting at 27 Indonesian stations

Experimental model-development report — not yet an operational public product
04 September 2026

Scope. Daily 00 UTC forecasts at +3, +6, +12, +24, +48, and +72 hours. The output is predictive and does not provide causal or source attribution.

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1 Executive finding

The validation-selected model is **CAMS model-output-statistics LightGBM**. Across the six lead times in the untouched 1 January–31 August 2026 test period, its mean station-balanced mean absolute error (MAE) is **9.82 $\mu\text{g m}^{-3}$** , corresponding to **+28.5%** mean skill relative to persistence and **+34.2%** relative to raw CAMS. Improvement occurs in **94.4%** of station–lead combinations, so aggregate skill does not imply uniform local benefit. The nominal 80% prediction intervals attain **78.8%** mean test coverage across leads.

Operational decision. Retain the system as pre-operational. It is suitable for scheduled shadow forecasts and scorecard monitoring, but public or duty-forecaster reliance should wait for prospective evaluation, explicit input-latency tests, a second forecast cycle, and an assessment of forecast meteorology.

2 Question and intended use

The question is whether a pooled machine-learning model can improve station-level hourly $\text{PM}_{2.5}$ forecasts over simple persistence at 27 BMKG locations while remaining computationally light enough for routine operation. One direct model is fitted for each forecast lead; direct forecasting avoids recursive error propagation. The intended output is a concentration forecast in $\mu\text{g m}^{-3}$ plus a calibrated uncertainty interval at each observed station.

The primary inference domain is the existing network. A separate station-holdout experiment asks how well the approach transfers to a location absent from fitting. Neither experiment creates a continuous spatial concentration field.

3 Data, quality control, and provenance

The supplied archive contains **1,158,532** rows from 27 hourly station files spanning 2021-01-01 to 2026-08-31. After documented error screening, **1,116,030 (96.33%)** source rows contain valid $\text{PM}_{2.5}$. Exact duplicated station-hours were removed only in derived data; no conflicting duplicate key was accepted. Negative $\text{PM}_{2.5}$ and values at or above $985 \mu\text{g m}^{-3}$ were treated as instrument error values. Plausible extremes below the cutoff were retained.

Relative humidity outside (0,100]% and temperature outside -10 to 50 °C were made unavailable as predictors without altering $\text{PM}_{2.5}$. This screening identified a block of physically impossible temperature values at Koto Tabang; the values were not corrected or imputed. Median station-level valid $\text{PM}_{2.5}$ coverage within each station’s observed span is **94.3%**.

CAMS global atmospheric-composition forecast $\text{PM}_{2.5}$ was extracted from the official ECMWF/CAMS/NRT Earth Engine mirror [7] and bilinearly sampled to station coordinates. Source mass density in kg m^{-3} was converted explicitly to $\mu\text{g m}^{-3}$. The parent CAMS service provides global forecasts twice daily on a 0.4° grid and is an atmospheric model product, not a surface observation [1]. The mirror is three-hourly, so +3 h is the earliest evaluated lead; no +1 h field was synthesized.

The extract retained **215,916 of 216,918 station–issue–lead records (99.54%)**. Missing records comprise a partial spatial gap on 17–23 February 2024 and a fully missing 00 UTC initialization on 30 June 2025; they were not imputed. A direct Copernicus archive check covered 20 in-domain stations and five common leads on 1 January 2026. Across 100 native grid-point comparisons, the maximum absolute difference was $0.000017 \mu\text{g m}^{-3}$; station-resampling implementations differed by $0.107 \mu\text{g m}^{-3}$ on average and $1.217 \mu\text{g m}^{-3}$ at most. This supports identity of the underlying field while documenting interpolation-level differences. The bounded experiment uses CAMS $\text{PM}_{2.5}$ plus observed temperature and humidity histories; forecast meteorology is a declared next-stage input.

Observation availability is heterogeneous across the 27-station network

Median valid PM_{2.5} coverage is 94.3% within each station's observed span; blank cells denote no station-month coverage.

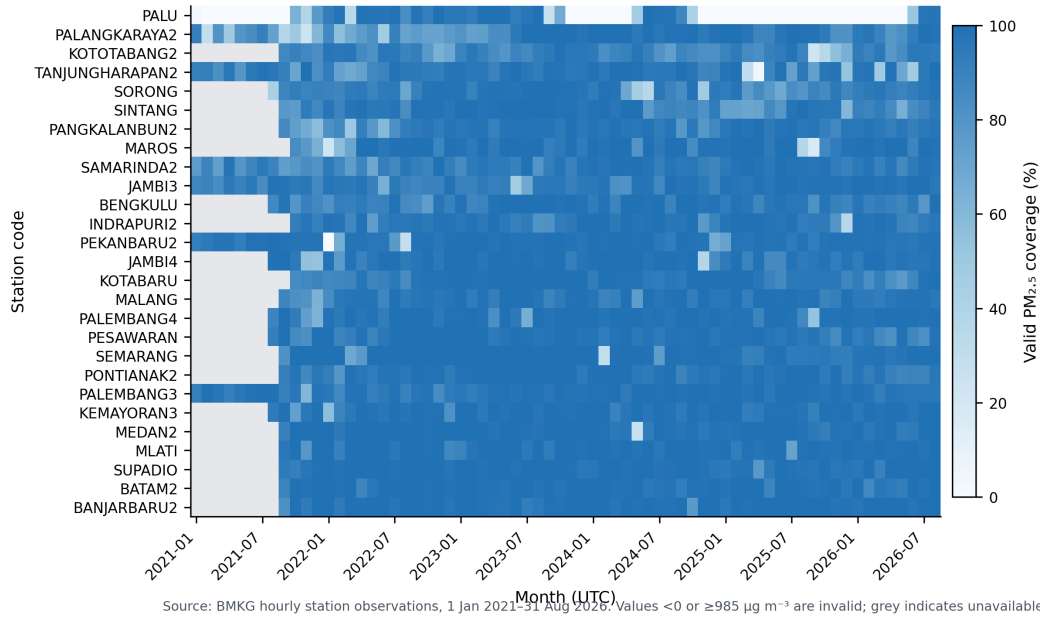


Figure 1: Monthly valid PM_{2.5} coverage across the 27 stations. Missing coverage is distinct from a valid zero concentration.

4 Prediction design

4.1 Inputs

Predictors use only timestamps at or before issue time: PM_{2.5} lags from 0 to 168 h; temperature and relative-humidity lags to 24 h; rolling PM_{2.5} mean, standard deviation, maximum, and availability over 3–168 h; local-hour and annual-cycle encodings; station coordinates and region; contemporaneous and distance-weighted network context within 400 km; and forecast-valid CAMS PM_{2.5}. Missing predictor values are retained for native tree handling. Feature importance is diagnostic and not causal importance.

4.2 Chronological separation

Targets from 2023–2024 form training, 2025 is used for point-model early stopping and model choice, and 2026 through 31 August is evaluated once as the final test. The uncertainty workflow adds a nested temporal separation: January–June 2025 tunes quantile tree counts, while July–December 2025 is reserved exclusively for conformal calibration. Splits are assigned by target time, not issue time. The derived table contains **216,918** station–issue–lead rows and **200,724** valid targets. Automated audits found 0 duplicate modeling keys, 0 non-future targets, and no target-time overlap among train, validation, and test. Chronological and station-blocked validation are used because random splitting of structured environmental data can underestimate predictive error [6].

4.3 Models and uncertainty

Persistence and a training-only station/local-time climatology are reference forecasts. Candidate learners are observation-only LightGBM, CAMS model-output-statistics (MOS) LightGBM, and CAMS MOS XGBoost. Gradient-boosted trees represent nonlinear interactions and missingness without the compute cost of deep sequence models [3, 4]. CAMS calibration with machine learning has prior scientific precedent, although performance is domain-specific [2].

Point and quantile predictions are clipped only at the physical lower bound of $0 \mu\text{g m}^{-3}$. Quantile LightGBM estimates the 10th, 50th, and 90th conditional percentiles. Quantile crossing is removed by sorting, then a lead-specific split-conformal expansion is estimated from the held July–December 2025 calibration block pooled across stations. Neither fitting nor interval calibration accesses 2026. Conformalized quantile regression provides a principled basis for adaptive prediction intervals under exchangeability assumptions [5]; temporal autocorrelation and later CAMS system changes mean empirical monitoring remains necessary.

5 Validation selection and independent test results

Model choice used mean station-balanced validation MAE across all six leads. The specified tie-break prefers CAMS LightGBM when it lies within 1% of the minimum, reducing deployment complexity while retaining a physically based forecast input.

Candidate	Validation MAE	Skill vs persistence (%)	Selected
CAMS MOS LightGBM	7.12	28.7	Yes
CAMS MOS XGBoost	7.16	28.0	No
Observation-only LightGBM	7.19	28.3	No

The selected model’s validation criterion is **7.12 $\mu\text{g m}^{-3}$** . The following table reports the subsequently opened test set. Positive bias means overprediction. The confidence interval is a 1,000-replicate station-week block bootstrap for absolute-error improvement over persistence.

Lead (h)	n	MAE	Persistence MAE	Raw CAMS MAE	Skill (%)	RMSE	Bias	r	95% CI: MAE gain
+3	5,918	7.62	12.64	9.95	36.5	16.48	-2.36	0.73	[3.79, 6.29]
+6	5,753	6.29	14.18	7.91	48.0	13.44	-1.58	0.72	[7.26, 9.59]
+12	5,897	10.50	16.69	17.61	32.9	16.62	-2.23	0.58	[5.77, 8.14]
+24	5,859	10.84	13.11	17.51	17.6	19.55	-3.43	0.66	[1.02, 2.45]
+48	5,839	11.66	13.90	18.10	16.9	21.39	-4.41	0.59	[0.99, 3.10]
+72	5,832	12.03	14.68	18.55	19.2	21.83	-4.45	0.57	[1.16, 3.50]

Forecast error increases with lead time, but learned models retain skill

The validation-selected CAMS MOS LightGBM reduces station-balanced 72 h MAE by 19.2% relative to persistence on the untouched test period.

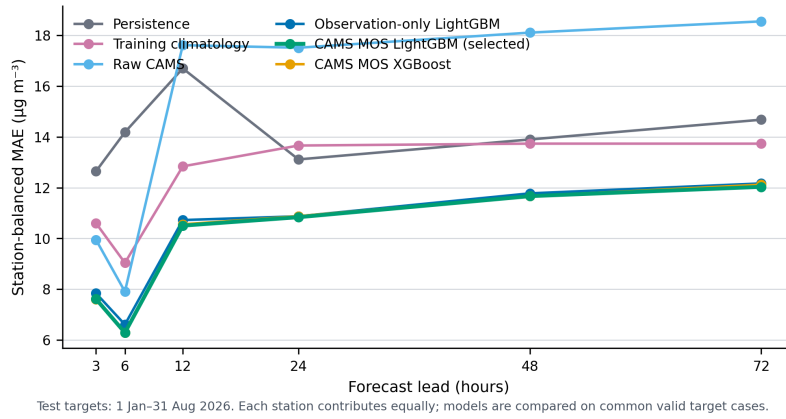


Figure 2: Station-balanced test MAE by lead for all candidates and reference forecasts.

Forecast skill varies materially by station and lead

The selected model improves on persistence in 94% of station-lead combinations; red cells identify operational failure modes requiring monitoring.

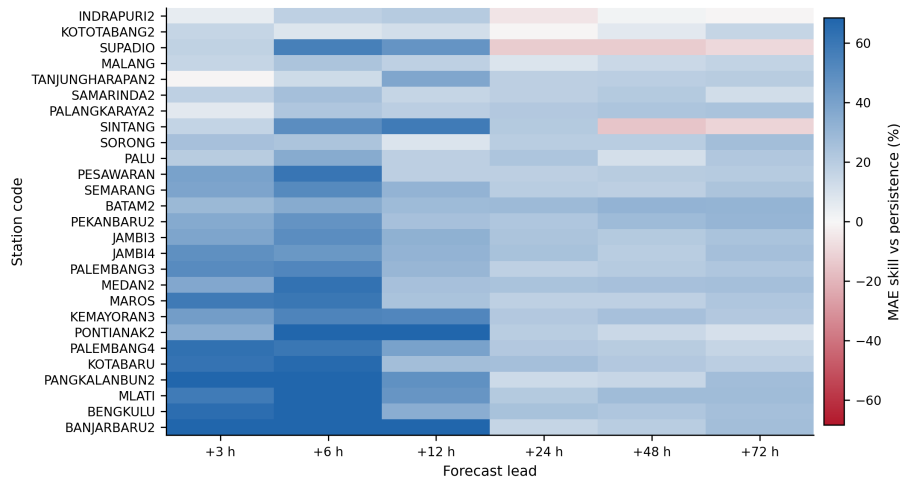


Figure 3: Selected-model MAE skill relative to persistence for every station and lead.

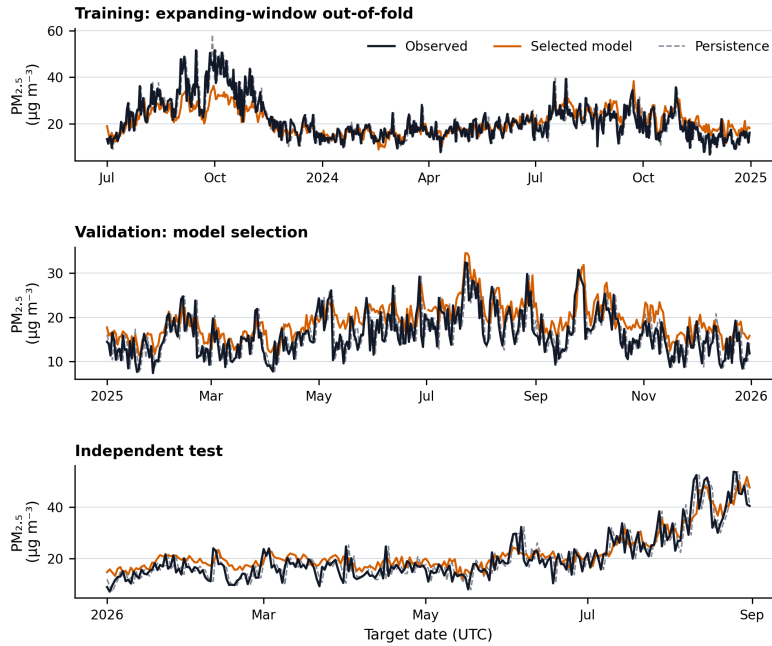
5.1 Chronological behaviour across development stages

Aggregate errors do not show whether the forecast follows episode timing, reacts late, or compresses peaks. Figure 4 therefore compares observed and predicted +24 h sequences in target-time order. Each point is the daily median across stations with valid data; no temporal smoothing or interpolation is applied.

Training-period values are not fitted predictions. They use expanding-window assessment blocks for July–December 2023, January–June 2024, and July–December 2024. Every assessment target is later than all targets used to fit its fold. The displayed model family and tree counts were selected later using 2025 validation, so this retrospective out-of-fold diagnostic describes temporal behaviour and is not an independent model-selection score. The 2025 panel is validation evidence and the 2026 panel remains the independent test.

Chronological evaluation exposes timing and peak errors hidden by aggregate scores

Daily station-median +24 h forecasts are shown without smoothing; training-period values are expanding-window out-of-fold predictions, not fitted values.



Network median across available stations for the 00 UTC cycle; one target per station-day. Missing calendar days remain gaps. Model family and tree counts were selected using 2025 validation.

Figure 4: Chronological observed, selected-model, and persistence PM_{2.5} at +24 h for expanding-window out-of-fold training assessments, validation, and independent testing. Values are daily station medians for the 00 UTC forecast cycle.

An accompanying nine-page station atlas shows the complete independent-test sequences at +6, +24, and +72 h for all 27 stations, including calibrated 10th–90th percentile intervals. It is supplied separately as a vector PDF so poor-performing stations and short-lived episodes remain inspectable rather than concealed by national aggregation.

5.2 Incremental value of CAMS

Relative to the otherwise identical observation-only LightGBM, adding forecast-valid CAMS PM_{2.5} reduced station-week-balanced MAE by **0.065 µg m⁻³** in validation (95% bootstrap interval 0.034 to 0.098) and **0.178 µg m⁻³** in the independent test (95% interval 0.119 to 0.242). This is an aggregate predictive association, not causal evidence. Test-period gains are clearest at +3 to +12 h; intervals cross zero at each of +24, +48, and +72 h, so longer-lead incremental benefit remains inconclusive individually.

Lead (h)	MAE gain	Relative gain (%)	95% CI	Station-weeks
+3	0.295	3.34	[0.194, 0.394]	945
+6	0.390	5.43	[0.286, 0.489]	944
+12	0.240	2.17	[0.165, 0.315]	944
+24	0.012	0.11	[-0.179, 0.184]	945
+48	0.045	0.36	[-0.057, 0.143]	945
+72	0.089	0.68	[-0.033, 0.222]	945
All leads	0.178	1.65	[0.119, 0.242]	945

The five least favourable station–lead combinations include SINTANG (-15.2%), SUPADIO (-13.1%), SUPADIO (-12.9%), SINTANG (-10.2%), SUPADIO (-9.5%). These are failure modes for investigation, not grounds for deleting observations or selectively omitting stations.

The selected model captures the central range but compresses some extremes

Hexagonal density reveals the full test distribution; the dashed identity line marks an unbiased point forecast.

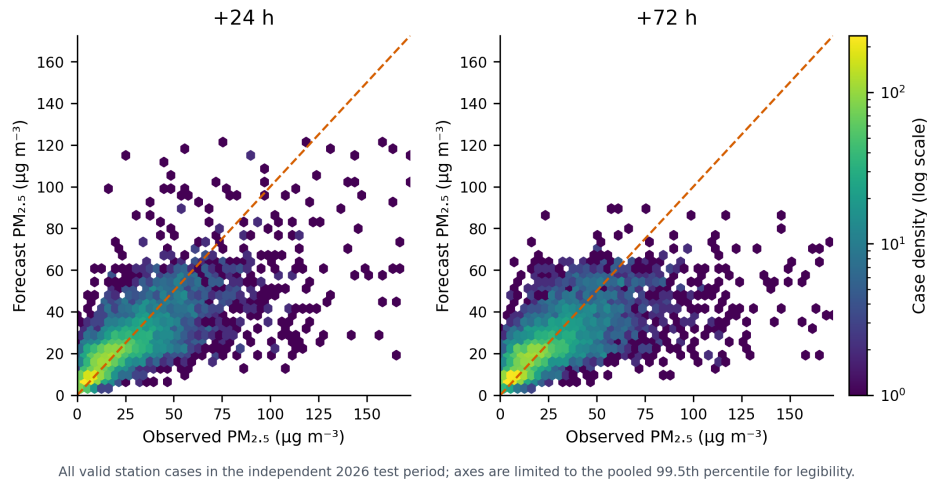


Figure 5: Observed versus selected-model PM2.5 at +24 h and +72 h. Display axes are limited to the pooled 99.5th percentile for legibility; metrics use the full valid range.

6 Uncertainty and high-concentration performance

Lead (h)	n	Coverage (%)	Mean width	Interval score
+3	6,050	79.4	19.8	42.3
+6	5,905	79.3	17.7	33.0
+12	6,108	81.1	34.0	53.0
+24	6,084	78.0	31.7	53.4
+48	6,084	77.5	32.3	57.6
+72	6,084	77.5	32.5	60.6

Prediction intervals quantify uncertainty, with coverage checked independently

Nominal 80% intervals achieve 78.8% mean empirical coverage across leads in the untouched test period.

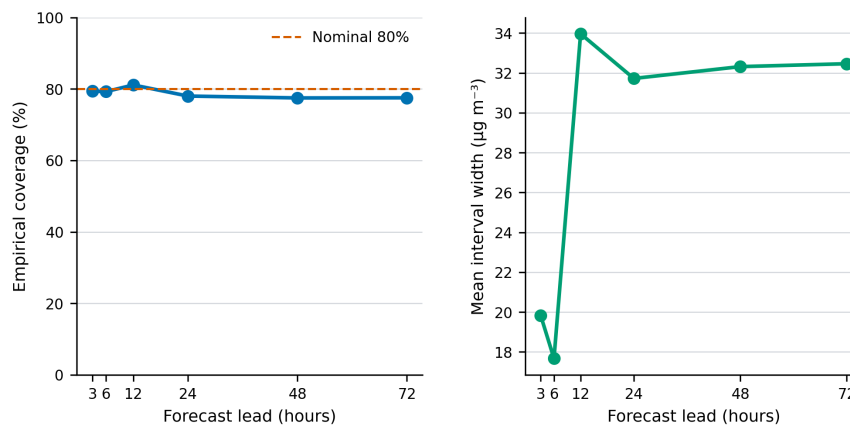


Figure 6: Independent empirical coverage and mean width of the nominal 80 percent intervals.

High-concentration events are defined separately for each station and lead using the station’s training-period 90th percentile. Probability of detection (POD) is the fraction of observed events detected; false-alarm ratio (FAR) is the fraction of predicted events that did not occur; critical success index (CSI) penalizes misses and false alarms.

Lead (h)	Hits	Misses	False alarms	POD (%)	FAR (%)	CSI (%)
+3	392	322	148	54.9	27.4	45.5
+6	403	314	125	56.2	23.7	47.9
+12	225	475	85	32.1	27.4	28.7
+24	357	423	148	45.8	29.3	38.5
+48	280	500	153	35.9	35.3	30.0
+72	252	528	154	32.3	37.9	27.0

High-concentration episodes remain the hardest operational regime

Mean critical success index is 36.2% for events defined independently from training-period station 90th percentiles.

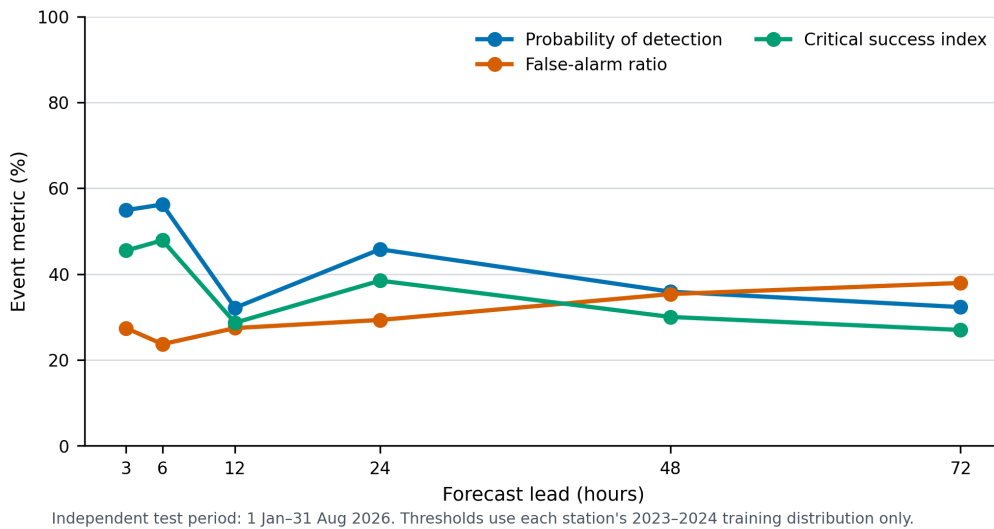


Figure 7: High-concentration event performance on the independent test period.

7 Robustness, spatial transfer, and interpretation

The same-network test estimates performance for stations represented in training. Five station folds provide a separate transfer diagnostic: each held station is tested with a model fitted to the remaining stations and without station identity. Fold assignment is independent of target values.

Lead (h)	Stations	MAE	Skill vs persistence (%)
+3	27	7.78	35.0
+6	27	6.50	46.4
+12	27	11.29	25.6
+24	27	11.09	14.1
+48	27	11.98	13.0
+72	27	12.42	14.4

Transfer to stations excluded from fitting is weaker than same-network prediction

Five-fold station holdout yields +24.7% mean MAE skill versus persistence across leads, without station identity as a feature.

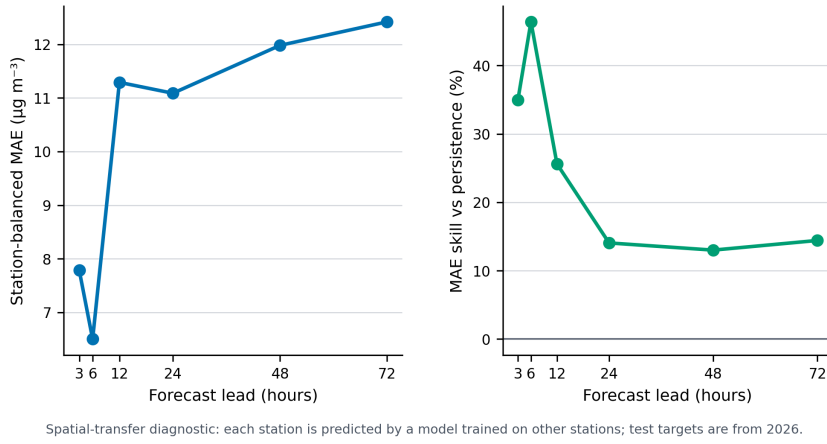


Figure 8: Performance when test stations are excluded from model fitting.

The 2024-only training sensitivity holds model form and tree counts fixed, changing only the fitting window. A positive difference means the shorter window has higher test MAE.

Lead (h)	Frozen MAE	2024-only MAE	Difference
+3	7.62	7.85	0.24
+6	6.29	6.63	0.34
+12	10.50	10.99	0.49
+24	10.84	11.28	0.44
+48	11.66	12.01	0.35
+72	12.03	12.39	0.36

Distribution-shift diagnostics compare predictor missingness, standardized mean differences, and population stability index between training and test. These statistics identify monitoring candidates; they do not establish that a shift caused an error. Residual bias is stratified by station, target month, and local target hour.

Forecast skill is distributed across persistence, CAMS, time, and site context

cams_pm25_ug_m3 has the largest mean tree importance across leads; importance is predictive and must not be read causally.

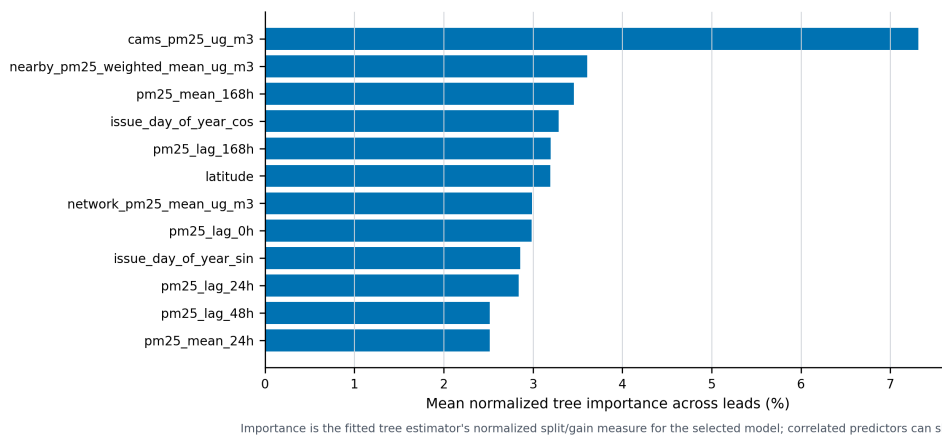


Figure 9: Mean fitted-tree feature importance for the selected model. Correlated predictors can exchange importance.

Residual bias changes with month and forecast lead

Positive cells indicate overprediction and negative cells underprediction; seasonal structure is a monitoring trigger, not proof of a physical cause.

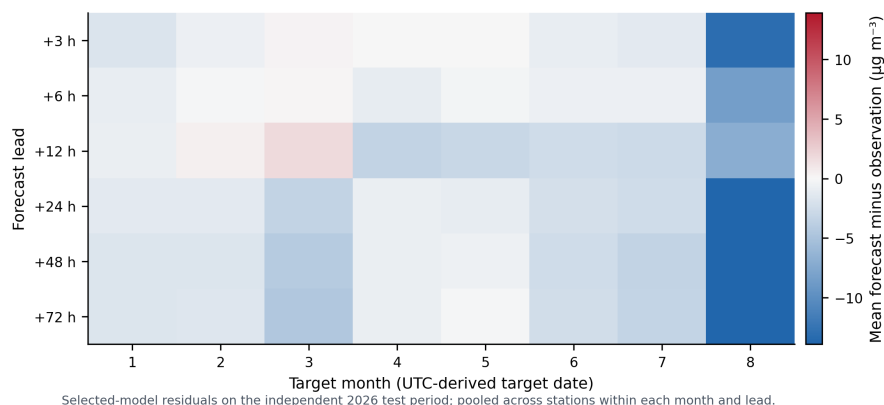


Figure 10: Selected-model mean residual by target month and lead.

8 Computational requirements and operational design

The experiment is designed for a CPU workstation; no GPU is required. Training uses at most 8 CPU threads. Measured peak memory was **preparation 1.43 GiB, training 0.81 GiB**. Measured preparation time was **92.9 s**, candidate/uncertainty/transfer evaluation was **103.5 s**, and deployment refitting was **24.4 s**. The complete deterministic pipeline took **4.7 min** across its latest successful stage measurements. The serialized deployment bundle is **9.8 MiB**. Warm inference across all point, fallback, and interval models totals about **0.136 s**, excluding model loading, input download, and feature preparation. The measured prepared-input end-to-end command took **2.13 s** for 162 station–lead rows.

For a routine 00 UTC run, a practical allocation is **4–8 CPU cores, 4 GiB RAM, and 2 GiB working storage**. Model inference should finish in under one minute; CAMS acquisition is network- and service-limited and should be budgeted at **5–30 minutes** with retries. The initial 3.7-year Earth Engine backfill required **12.7 min** of measured server-query time. A full deterministic research rebuild should be budgeted at **5–15 minutes** on a comparable CPU workstation; slower CPUs and uncached report dependencies may require longer.

Deployment point models are refitted through December 2025; deployment quantile models stop at June 2025 so July–December remains a held calibration block. If CAMS PM_{2.5} is missing, a separately fitted observation-only point forecast is used and the status is marked degraded; calibrated intervals are withheld. Observation older than six hours is explicitly flagged. This fallback supports continuity but must not be presented as equivalent quality.

8.1 Implemented daily shadow workflow

The non-public shadow workflow is scheduled daily at 17:15 WIB (10:15 UTC). It saves an immutable BMKG dashboard snapshot, freezes the observation cutoff, acquires the current CAMS 00 UTC initialization directly from the Copernicus archive, writes atomic forecasts with hashes and freshness/status fields, and scores them only after observations appear. First-seen station-hour values are retained for verification and later raw snapshots preserve revisions. There is no public upload.

The first end-to-end engineering run on 4 September 2026 generated **162 rows in 67.7 s** with **0 degraded rows**. It produced **108 prospectively eligible rows** and labelled **54 rows** whose target times had already occurred. This first execution is a workflow test, not a prospective performance result.

That engineering run completed **11.3 h after the 00 UTC initialization**. Observation freshness in shadow

output is evaluated relative to actual generation time as well as model initialization; otherwise an observation timestamped at 00 UTC would be incorrectly described as fresh about eleven hours later.

1. Retrieve and validate the 00 UTC CAMS forecast after publication.
2. Freeze the latest observation cutoff and record station freshness.
3. Construct predictors without accessing later observations.
4. Produce forecasts, intervals, status flags, hashes, and logs.
5. Score forecasts when observations arrive; retain missing cases rather than silently backfilling.
6. Record warnings for missing CAMS, stale station data, schema/version changes, degraded operation, extreme residuals, and interval undercoverage.

9 Limitations and readiness gates

- One daily cycle was retrospectively tested; this is not evidence for 12 UTC or arbitrary issue times.
- Direct CAMS availability is later than model initialization. At the installed schedule, +3 h and +6 h are latency diagnostics rather than prospective forecasts; +12 h is eligible only if generation completes before 12 UTC. Prospective reporting uses the stored per-row eligibility flag.
- The test covers eight months of 2026, not a complete annual cycle, and no prospective shadow period has yet been observed.
- CAMS is much coarser than a station and its forecast system can change; a sampled grid value is not a station measurement.
- The CAMS mirror is 99.54% complete for the specified station–issue–lead grid; missing cycles are excluded from primary complete-case comparisons and require the declared degraded fallback in operations.
- Forecast meteorology was not included. Observed meteorology at or before issue time cannot represent future dispersion conditions.
- Missingness, maintenance, relocation, calibration change, and reporting latency can alter station performance.
- Lead-specific conformal calibration pooled across stations assumes exchangeability more strongly than this autocorrelated, spatially dependent archive warrants. The later calibration block prevents fitting leakage, but held-out coverage is reported rather than guaranteed operationally.
- Station-transfer results do not validate a continuous spatial map or arbitrary new station.
- Feature importance is not emissions, wildfire, transport, contribution, or causal attribution.

Before public operations, require at least 60–90 days of prospective shadow scoring; a documented data-arrival cutoff; 12 UTC backtesting if desired; incremental-skill tests for forecast wind, boundary-layer height, humidity, and precipitation; CAMS-version monitoring; station-specific fallback rules; alerting and rollback; and scientific sign-off on thresholds and wording.

10 Reproducibility and data-quality detail

Every source observation file, external archive, sampled field, configuration, research model, and deployment model has a SHA-256 checksum in the provenance record. Random seeds are fixed. Derived tables preserve UTC timestamps, source station identity, explicit units, target lead, split, and missingness. No source observation was changed.

Station	Name	Start	End	Valid coverage (%)	Absent hours	Invalid T
PALU	Lore Lindu Bariri	2021-01-23	2026-08-31	42.6	25,203	0
PALANGKARAYA2	Palangka Raya	2021-01-01	2026-08-31	84.6	4,564	0
KOTOTABANG2	Koto Tabang	2021-09-24	2026-08-31	85.7	1,805	3,662
TANJUNGHARAPAN2	Tanjung Harapan	2021-01-01	2026-08-31	88.1	4,513	0
SORONG	Sorong	2021-08-19	2026-08-31	88.9	2,250	0
SINTANG	Sintang	2021-09-19	2026-08-31	91.0	496	0
PANGKALANBUN2	Pangkalan Bun	2021-09-24	2026-08-31	91.2	1,353	0
MAROS	Maros	2021-10-13	2026-08-31	91.3	2,070	0
SAMARINDA2	Samarinda	2021-01-01	2026-08-31	91.8	1,181	0
JAMBI3	Kota Jambi	2021-01-01	2026-08-31	92.8	2,479	0
BENGKULU	Bengkulu	2021-08-19	2026-08-31	93.4	576	0
INDRAPURI2	Indrapuri	2021-10-04	2026-08-31	93.6	757	0
PEKANBARU2	Pekanbaru	2021-01-01	2026-08-31	93.6	2,547	0
JAMBI4	Muaro Jambi	2021-08-18	2026-08-31	94.3	1,742	0
KOTABARU	Kotabaru	2021-10-09	2026-08-31	94.7	52	0
MALANG	Malang	2021-09-23	2026-08-31	94.8	1,085	0
PALEMBANG4	Musi 2 Palembang	2021-08-21	2026-08-31	95.6	1,191	0
PESAWARAN	Pesawaran	2021-08-05	2026-08-31	95.8	890	0
SEMARANG	Semarang	2021-09-02	2026-08-31	95.8	1,609	0
PONTIANAK2	Mempawah	2021-09-09	2026-08-31	96.3	629	0
PALEMBANG3	Talang Betutu Palembang	2021-01-04	2026-08-31	96.4	928	0
KEMAYORAN3	Kemayoran	2021-08-25	2026-08-31	96.6	785	0
MEDAN2	Medan	2021-09-01	2026-08-31	97.1	946	0
MLATI	Sleman	2021-09-09	2026-08-31	97.4	286	0
SUPADIO	Kubu Raya	2021-09-09	2026-08-31	97.9	456	0
BATAM2	Batam	2021-09-14	2026-08-31	97.9	292	0
BANJARBARU2	Banjarbaru	2021-09-08	2026-08-31	98.2	234	0

11 References

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